Khwaja Yunus Ali University

School of Science and Engineering

B.Sc. in CSE & EEE

Lesson Plan

Course Title : Physics - II
Course Code : CSE 0533 1201 (CSE) & PHY 0533 1201 (EEE)

Term: Spring – 2024	Date: 15.01.2024	Time: 03.30 PM	Room: 327	Activity: Lecture 5

Topics to be discussed: Applications of the First Law of Thermodynamics
Important definitions, different thermodynamical processes, relation between C_v, C_p & PVT, Problem solving

Target CLO & PLO : CLO2-3 & PLO2-5

Learning Outcomes : At the end of the class the students will be able to:

- ❖ Define C_p and C_v.
- ❖ Explain Adiabatic & Isothermal Prosess.
- ❖ Define γ
- ❖ Establish the relation C_p & C_v also P, V & T.
- ❖ Solve assigned recommended mathematical problems

Methods of Teaching:

Mode/Activities	Time
Feedback of previous class	10min
Interactive discussion on some natural phenomenon	10min
Lecture Presentation on Topics	30 min
Group activity	25 min
Feedback session	10 min

Assessment Strategy : Quizzes/Class tests, Class Participation

Sample Questions (MCQ):

1) What is the unit of γ?
a) N b) m c) F d) Nm^{-1} e) unitless
2) Internal energy depends only on the –
a) pressure b) volume c) elasticity d) temperature e) mass
3) Which one is correct?
a) $C_p - C_v = \gamma$ b) $C_p - C_v = R$ c) $C_v - C_p = \gamma$ d) $C_v - C_p = R$ e) $C_v/C_p = \gamma$

Sample Questions (SAQ):

1) Distinguish between isothermal and adiabatics processes.
2) Derive the relation between Cp and Cv. Here the symbols have their normal meaning.
3) Or, show that, for an ideal gas Cp is always greater than Cv.
4) What is γ in thermodynamics?
5) Derive the Relation between P, V & T of a gas in Adiabatic Change.
6) Or, Derive the Relation $PV^{\gamma} = Constant$. from first law of thermodynamics.
7) Or, Establish the relation between pressure and volume in adiabatics process.
8) Or, Find the relation between volume and temperature in adiabatics.
9) Show that the adiabatic curve is more steeper than the isothermal curve.
10) For an ideal gas $\gamma = 1.4$, calculate the values of molar specific heats of the gas. ($R = 8.31\ \mathrm{Jmol^{-1}K^{-1}}$)
11) The volume of air is made three times the original volume by adiabatic expansion. If the initial pressure is 1 atmospheric pressure, what is the final pressure? [$\gamma = 1.40$]
12) A fixed amount of gas at 270C is suddenly expanded to double its original volume. What is the final Temperature? [Name of the gas is Helium].

Reference Books: 1. Giasuddin ahmed – Physics for Engineers (Part-1)
2. Halliday Resnick - Fundamental of Physics, 2nd edition

Pre-reading book for next Lecture:

- Md. Ziaul Ahsan – Physics for Engineering (Lecture Series), Pages: 01-64
- Dr. Amir Hussain Khan – Physics 2nd Paper (HSC Text Book), Pages: 01-36